



Sanjivani Rural Education Society's
Sanjivani College of Engineering, Kopargaon-423 603
(An Autonomous Institute, Affiliated to Savitribai Phule Pune University, Pune)
NAAC 'A' Grade Accredited, ISO 9001:2015 Certified

Department of Computer Engineering

(NBA Accredited)

Subject- Object Oriented Programming (CO212)
Unit 3 – Polymorphism and Virtual Function
Topic – 3.1 Polymorphism



POLYMORPHISM

- ❖ The process of representing one thing in multiple forms is known as Polymorphism.
- ❖ Here one form represent original form or original method always resides in base class and multiple forms represents overridden method which resides in derived classes.
- ❖ Polymorphism word is derived from 2 Greek words: Poly and Morphs.
- ❖ The word "Poly" means many and "Morphs" means forms.

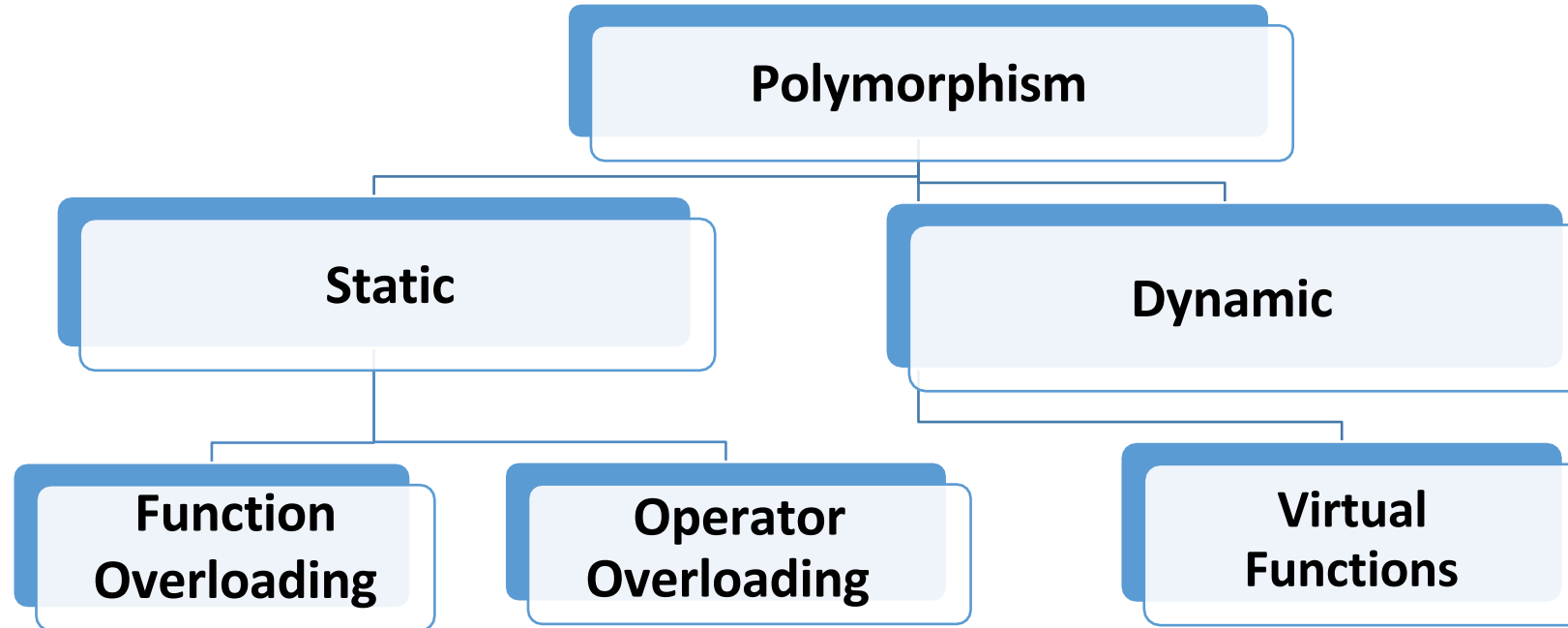


Real life example of Polymorphism

- ❖ Suppose if you are in class room that time you behave like a student, when you are in market at that time you behave like a customer, when you at your home at that time you behave like a son or daughter, Here one person have different-different behaviors.



Types of Polymorphism



- ✓ Static polymorphism is also known as early binding or compile-time polymorphism. In static polymorphism memory will be allocated at compile-time.
- ✓ Dynamic polymorphism is also known as late binding or run-time polymorphism. In dynamic polymorphism memory will be allocated at run-time.



Function Overloading:

- ❖ Whenever same function name is used multiple times in the same class with different number of parameters or different types of parameters is known as function overloading.

Operator Overloading:

- ❖ The process of making an operator to exhibit different behaviors in different instances is known as operator overloading.

Types Of Operator Overloading

- ❖ Unary operator overloading:

These operators have only single operand. e.g.++,--

- ❖ Binary operator overloading:

These operators can have two operands.

e.g. +,-*,/,%,^,,==,+=,&



Virtual Function

- ❖ A virtual function is a member function that is declared as virtual within a base class and redefined by a derived class.
- ❖ To create virtual function, precede the base version of function's declaration with the keyword virtual.
- ❖ Here we use a pointer to the base class to refer to all the derived objects.
- ❖ The method name and type signature should be same for both base and derived version of function.



Rules for using Virtual Function

- Virtual function cannot be static and also cannot be a friend function of another class.
- Virtual function should be accessed using pointer or reference of base class type to achieve run time polymorphism.
- The prototype of virtual function should be same in base as well as derived class.
- It is always defined in base class and overridden in derived class.
- A class may have virtual destructor but it cannot have a virtual constructor.



Virtual Function Example

```
class A {  
    public:  
        virtual void show()  
        {  
            cout<<"Content of base class.";  
        }  
};  
class B : public A  
{  
    public:  
        void show()  
        {  
            cout<<"Content of derived class.";  
        }  
};
```

```
int main()  
{  
    A b,*bptr; //Base class pointer  
    B d; //Derived class object  
    bptr = &b;  
    bptr->show(); //Dynamic Binding Occurs  
    bptr=&d;  
    bptr->show(); //Dynamic Binding Occurs  
    return 0;  
}  
Output: Content of base class.  
        Content of derived class.
```



- [Runtime polymorphism](#) is achieved in a case of Function Overriding.
- [Function overriding](#) occurs when a derived class has a same definition for one of the member functions of the base class. That base function is said to be overridden.

Thank You..